(전자 2 – B08)

Feature Channel Restoration and Average Pooling-Based Channel Resize for FCM

2026. 6. 16(화)

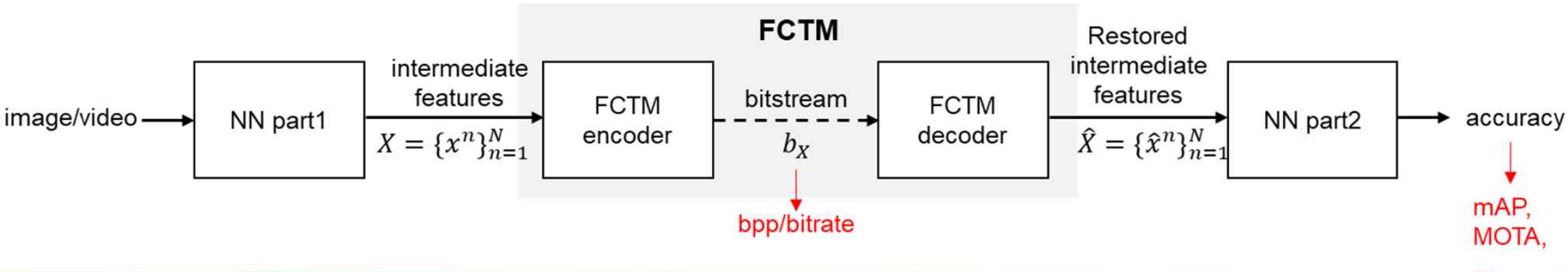
방준혁, 김민

한국항공대학교
Korea Aerospace University

Contents

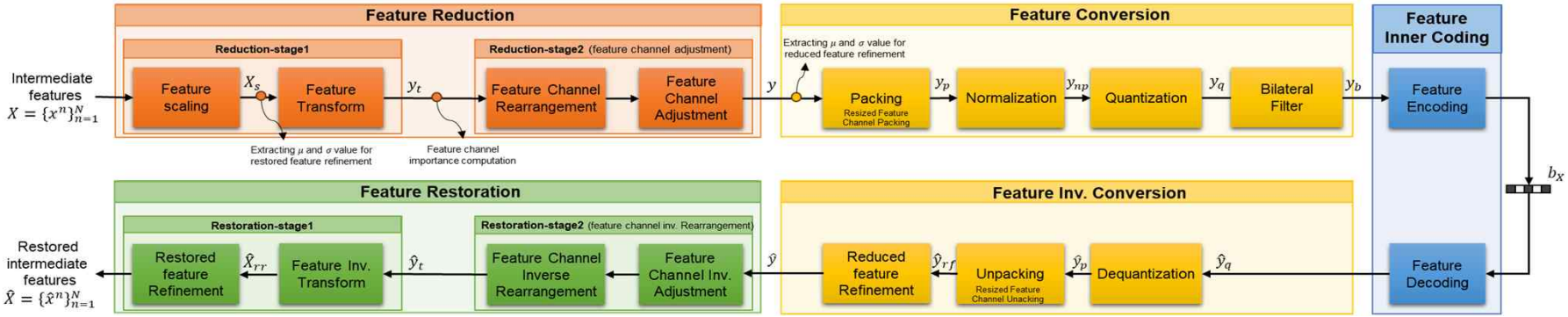
1. Introduction to FCM

- Feature Coding for Machines (FCM) in MPEG
  - Machine vision applications are rapidly expanding across various AI-driven domains, demanding efficient feature transmission under bandwidth constraints
  - As AI models grow larger (e.g., Vision Transformers, large-scale detection models), transmitting raw intermediate features becomes increasingly costly
    - Compression of intermediate features is critical for real-time inference in bandwidth-limited environments
  - MPEG is developing a standard for image/video feature compression to support machine vision tasks called Feature Coding for Machines (FCM)
    - Feature Compression Test Model (FCTM) to evaluate standard candidate technologies
      - FCTM 1 (Oct. 2023, Hannover) → FCTM 10 (Jan. 2026, Online)



2. Overall Framework of FCTM

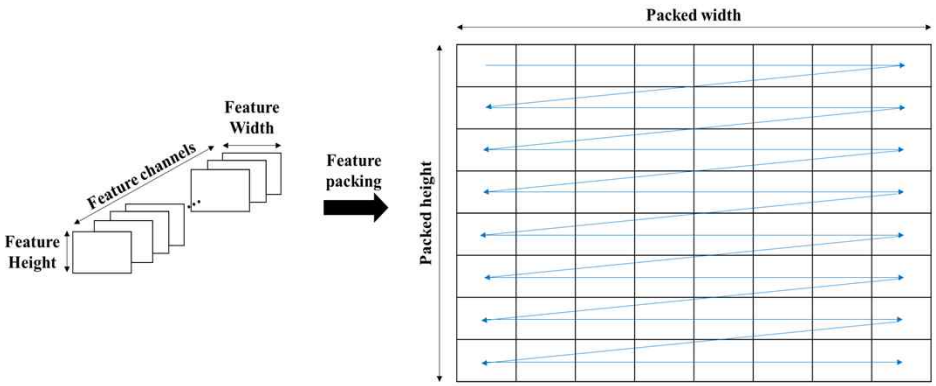
- Feature reduction
  - Multi-scale intermediate features are transformed into a single-scale representation via FENet with channel adjustment
- Feature conversion
  - Reduced features are packed into a 2D frame, followed by normalization and quantization for inner codec input
- Feature inner coding
  - Quantized feature frames are encoded/decoded using VVC codec
- Feature inverse conversion, feature restoration
  - Inverse operation of feature conversion and reduction



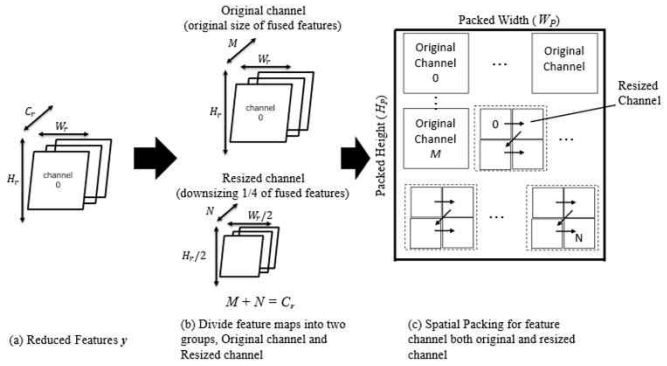
Overall framework of FCTM v10

Feature Channel Packing

- Default feature packing
  - Converts reduced feature y (3D tensor$[C_r, H_t, W_t]$) into a single 2D frame$[1, H_p, W_p]$ channels are arranged in raster scan order on the 2D frame
  - Number of horizontal/vertical channels determined by $long\_edge / short\_edge$
- Resized feature channel packing
  - Downscales less important channels to $1/4$ size before packing
    - 4 resized channels are merged into 1 channel:$C_r' = M + [N/4]$
    - Down-sampling method: Bilinear interpolation



Spatial packing for the feature channel



Pipeline of reseized feature packing

Removed Channel Restoration

- Feature channel restoration (Decoder)
  - Channels restored to original positions using received rearrangement index
    - Output tensor initialized with mean value $y_{m\ ean}$ of restored channels
    - Active channels → filled with restored values
    - Removed channels → replaced with $y_{m\ ean}$
  - Output: Restored Transformed Feature y_t, shape $[C_t, H_t, W_t]$
- Limitation of anchor method
  - Packing: resized channels down-sampled by bilinear interpolation
    - Simply reduces resolution without preserving structural information
  - Restoration: removed channels replaced with $y_{m\ ean}$ (mean of active channels)
    - No spatial or inter-channel correlation utilized

3. Proposed Method – Average Pooling-Based Channel Resize

- Existing method
  - Resized feature channels are down sampled using bilinear interpolation
  - Bilinear interpolation generates reduced features by sampling neighboring pixels with distance-based weights
  - For odd-sized feature maps, some spatial information is not reflected in the reduced feature representation due to non-uniform sampling locations
- Adaptive average pooling
  - Adaptive average pooling generates the target feature size through region-wise averaging
  - This approach provides stable feature resizing for odd-sized feature maps
  - Unlike bilinear interpolation, adaptive average pooling aggregates information from all input regions contributing to the target feature map

Average Pooling-Based Channel Resize

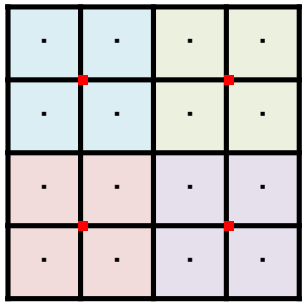
- Feature size analysis
  - Feature map sizes differ across datasets, and odd-sized dimensions are observed in SFU and PANDA
  - Resizing differences between bilinear interpolation and adaptive average pooling appear only for odd-sized feature maps

	Dataset	Feature Size	Type
Object Detection	SFU	$13 \times 20/13 \times 21/25 \times 40$	Odd Included
Object Tracking	TVD	12×20	Even x Even
	HiEve	12×20	Even x Even
Semantic Segmentation	PANDA	12×21	Even x Odd

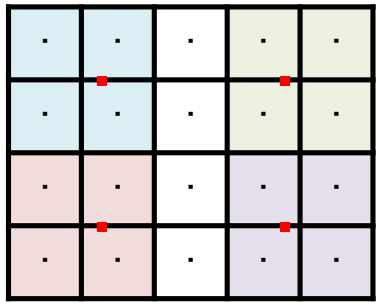
Size of single-scale feature channel

Average Pooling-Based Channel Resize

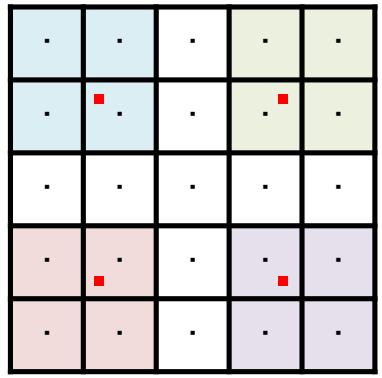
- Bilinear interpolation



Even x Even

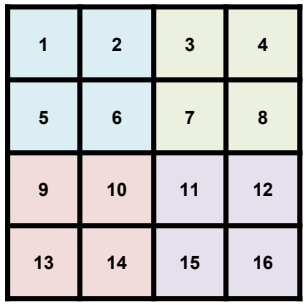


Even x Odd

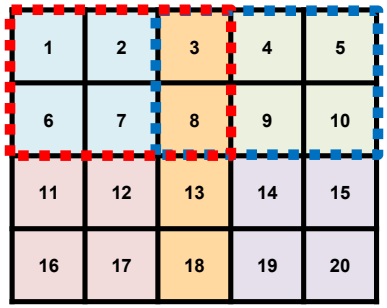


Odd x Odd

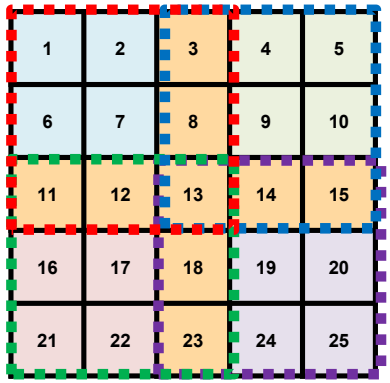
- Adaptive average pooling



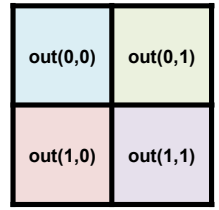
Even x Even



Even x Odd



Odd x Odd

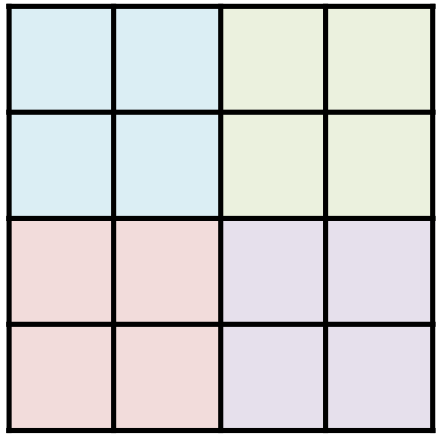


Output

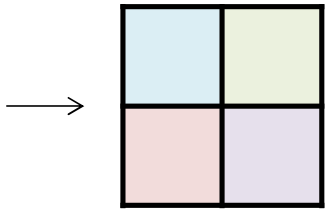
: overlap

Average Pooling-Based Channel Resize

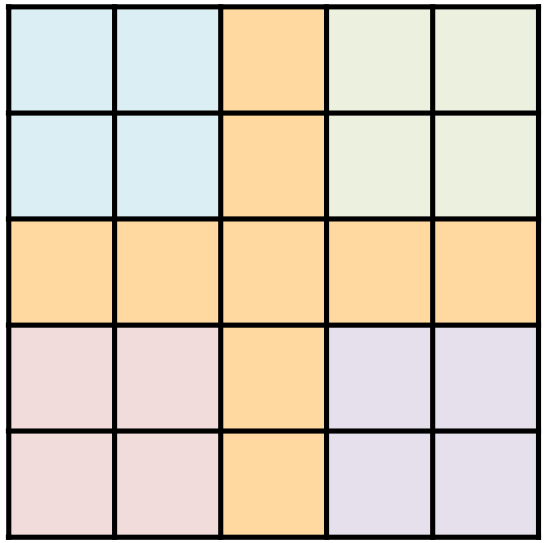
- Adaptive average pooling
  - Adaptive average pooling divides the input feature map into regions based on the target output size
  - The average value of each region is then used to generate the resized feature map



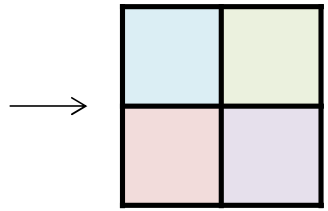
Input 4x4 (Tracking: 12x20)



Output 2x2 (Tracking: 6x10)



Input 5x5 (SFU: 13x21)



Output 2x2 (SFU: 6x10)

4. Active Channel Residual-Based Channel Restoration

- Anchor method
  - Removed channels are reconstructed using a scalar mean value
  - The same scalar mean is applied to all pixels in the removed channel
  - This approach is simple but does not reflect local spatial variations
- Active channel residual
  - A residual term is defined using neighboring active channels
  - The residual is computed as the pixel-wise difference between the right and left active channels
  - The restoration value is generated by the equation

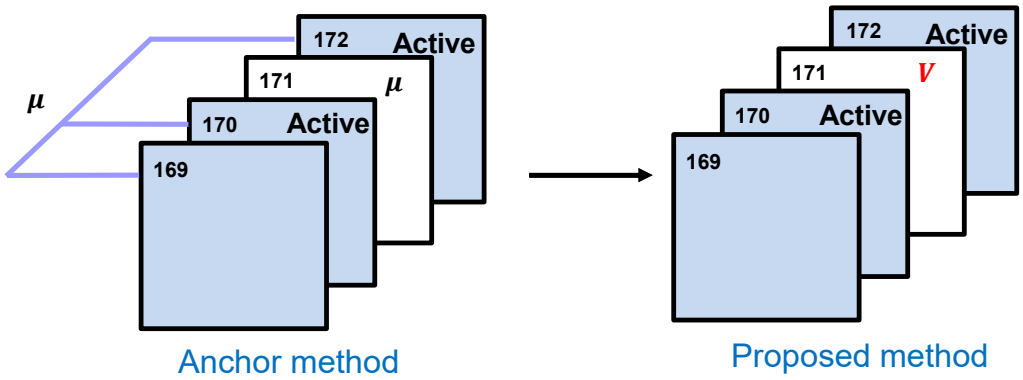
$$Removed\ Channel = \alpha \cdot (R - L)$$

$$\alpha = 0.05$$

R : $Right\ \ Active\ \ \ Channel\ \ Pixel\ \ Value$

L : $Left\ \ Active\ \ \ Channel\ \ Pixel\ \ Value$

α : $Scaling\ \ \ Factor$



Anchor method

Proposed method

5. Experimental Results

- Test conditions
  - The proposed methods were implemented in FCTM v10.0 and evaluated under the FCM CTTC
- Tasks
  - Object Detection: Locates objects with bounding boxes and classifies them
  - Object Tracking: Follows specific objects through video frames, maintaining their identity
  - Semantic Segmentation: Pinpoints objects' exact pixel boundaries

FCM Common Training and Test Conditions

Task	Dataset	Network	Split point	Rate measure	Task measure
Instance Segmentation	OpenImagesV6	MaskRCNN-X101-FPN	P-layer (P2-P5)	BPP	mAP @ 0.5
Object Detection	OpenImagesV6	FasterRCNN-X101-FPN	P-layer (P2-P5)	BPP	mAP @ 0.5
	SFU	FasterRCNN-X101-FPN	P-layer (P2-P5)	Kbps	mAP @ 0.5-0.95
Object Tracking	TVD	JDE-1088x608	Darknet-53	Kbps	MOTA
	HiEve	JDE-1088x608	Layers 75, 90, 105 (“ALT1”)	Kbps	MOTA
Semantic Segmentation	Pandaset	Panoptic-FPN-R101	P-layer (P2-P5)	Kbps	mIoU

Inference Environment

	Inference environment
OS	Ubuntu 22.04.4 LTS
CPU	AMD Ryzen Threadripper 7960X
Python	3.8.19
Numpy	1.24.4
Torch	2.4.1+cpu
Torchvision	0.15.1+cpu
CompressAi	1.2.9.dev0
CompressAi-Vision	1.1.14.dev0
ffmpeg	4.2.2

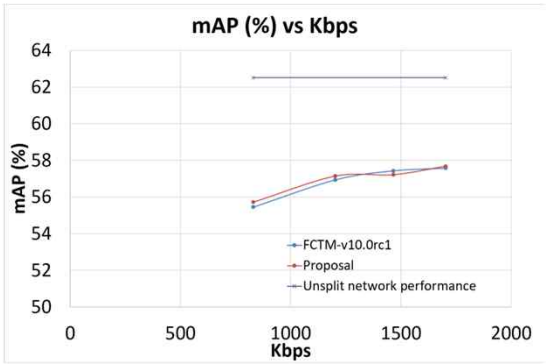
1) Resize Channel Pooling

- Experimental results – resize channel pooling
  - Comparison with the FCTM v10.0 anchor
  - The proposed method achieves an overall BD-rate reduction of -1.42%
  - No performance change was observed in object tracking tasks, where feature maps consist of even-sized dimensions (12×20)

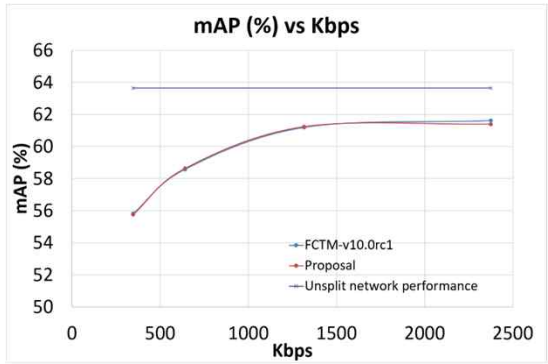
Proposal vs. FCTM-v10.0		
Task	Test Dataset	BD-rate [%]
Object Detection	SFU (Class A/B)	-6.37%
	SFU (Class C)	-1.76%
	SFU (Class D)	-4.06%
	SFU (Class A/B-no-rescale)	-3.98%
Object Tracking	TVD (OVERALL)	0.00%
	HiEve (1080p)	0.00%
	HiEve (720p)	0.00%
Semantic Segmentation	PANDAM1	1.48%
	PANDAM2	1.53%
	PANDAM3	-1.08%
	OVERALL	-1.42%

Resize Channel Pooling

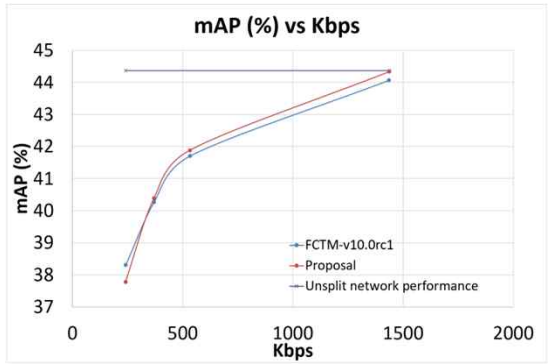
- RD-curves – resize channel pooling



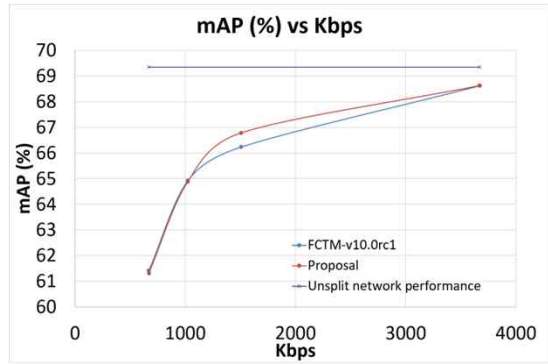
SFU Class A/B Detection



SFU Class C Detection



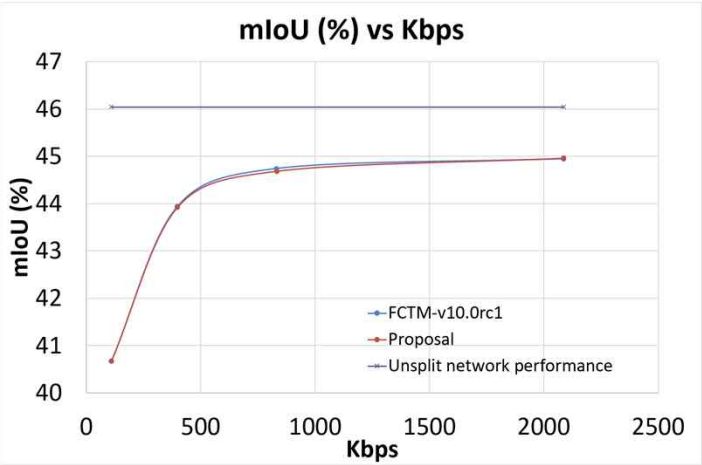
SFU Class D Detection



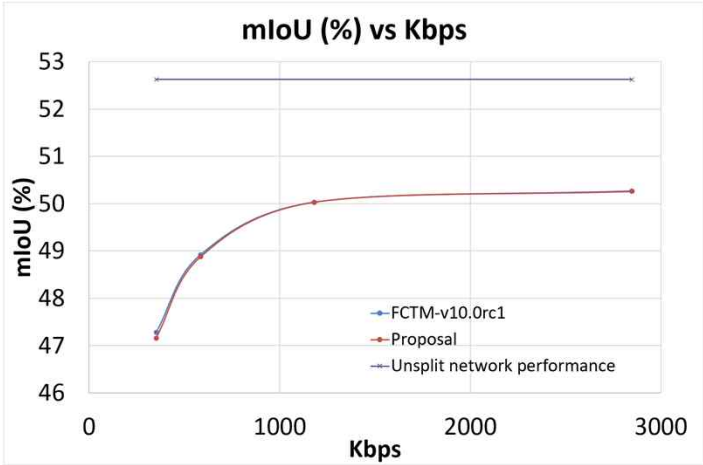
SFU Class A/B- no-rescale Detection

Resize Channel Pooling

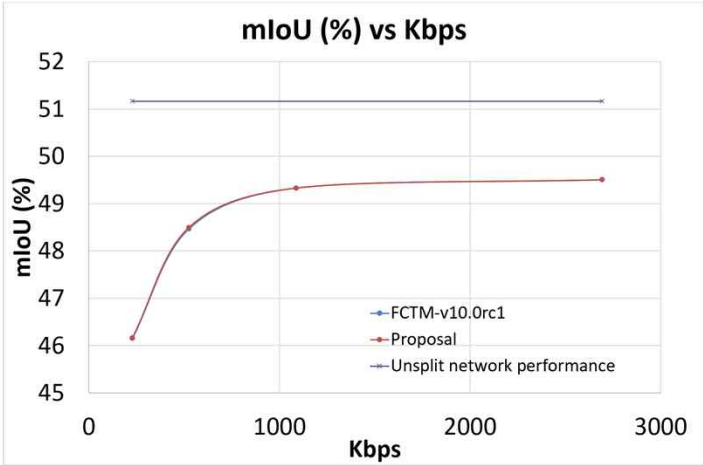
- RD-curves – resize channel pooling



PANDAM1 Segmentation



PANDAM2 Segmentation



PANDAM3 Segmentation

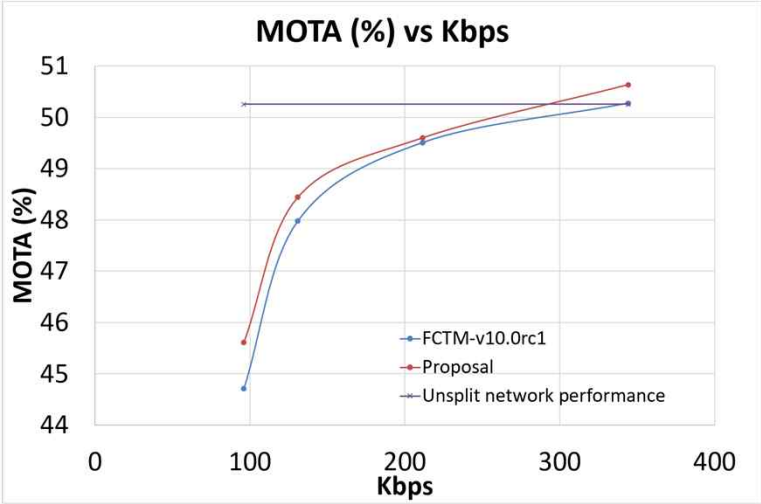
2) Active Channel Residual

- Experimental Results – Active channel residual
  - Comparison with the FCTM v10.0 anchor
  - The proposed method achieves an overall BD-rate reduction of -0.66%
  - Active channel residual showed mixed results across datasets, indicating that neighboring channel differences can affect restoration performance

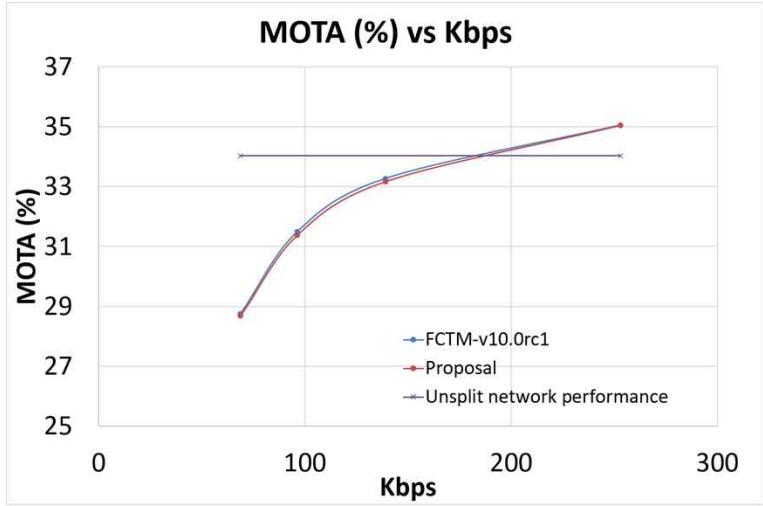
Proposal vs. FCTM-v10.0		
Task	Test Dataset	BD-rate [%]
Object Detection	SFU (Class A/B)	0.76%
	SFU (Class C)	-0.59%
	SFU (Class D)	-0.76%
	SFU (Class A/B-no-rescale)	-0.26%
Object Tracking	TVD (OVERALL)	-7.48%
	HiEve (1080p)	1.80%
	HiEve (720p)	-0.23%
Semantic Segmentation	PANDAM1	-0.04%
	PANDAM2	0.02%
	PANDAM3	0.16%
	OVERALL	-0.66%

Active Channel Residual

- RD-curves – active channel residual



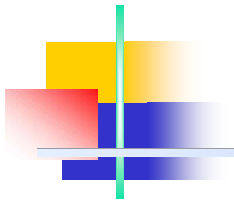
TVD overall Tracking



HiEve 1080p Tracking

6. Conclusions

- Proposed Methods for Enhancing FCTM Performance
  - Resize Channel Pooling
    - Replaced bilinear interpolation with adaptive average pooling in resize packing
    - Provides stable feature resizing for odd-sized feature maps
    - Utilizes the entire spatial region when generating resized features
  - Active Channel Residual
    - Replaced scalar mean with residual from neighboring active channels
    - Utilizes local differences between neighboring active channels
  - Conclusions
    - Both methods improve feature reconstruction without modifying the FCTM bitstream structure
    - Experimental results indicate the effectiveness of feature restoration techniques in FCTM
    - Resize channel processing and removed channel restoration can significantly affect the overall coding performance



Thank you for your attention!